

A Compact-Subordinate Stability Subcase for Quasi Norm Attainment in ℓ_∞ Range Sums

Run fa_banach_001, lane 13

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Source and targeted question

The source paper is Choi, Choi, Jung, and Martin, *On quasi norm attaining operators between Banach spaces*, arXiv:2004.11025. It defines $T \in \mathcal{L}(X, Y)$ to be quasi norm attaining if

$$\overline{T(B_X)} \cap \|T\|S_Y \neq \emptyset,$$

equivalently if some sequence $(x_n) \subset B_X$ satisfies $Tx_n \rightarrow y$ and $\|y\| = \|T\|$.

Problem 7.3 of the source asks:

Let X, Y_1, Y_2 be Banach spaces such that $\text{QNA}(X, Y_j)$ is dense in $\mathcal{L}(X, Y_j)$ for $j = 1, 2$. Is $\text{QNA}(X, Y_1 \oplus_\infty Y_2)$ dense in $\mathcal{L}(X, Y_1 \oplus_\infty Y_2)$?

subsection as they are related to results for the denseness of norm attaining operators.

Problem 7.3. Let X, Y_1, Y_2 be Banach spaces such that $\text{QNA}(X, Y_j)$ is dense in $\mathcal{L}(X, Y_j)$ for $j = 1, 2$. Is $\text{QNA}(X, Y_1 \oplus_\infty Y_2)$ dense in $\mathcal{L}(X, Y_1 \oplus_\infty Y_2)$?

The positive answer to this question for norm attaining operators was given in [50].

Let us comment that a positive answer to Problem 7.3 would give an example of a Banach space Z such that $\text{QNA}(Z, Z)$ is dense in $\mathcal{L}(Z, Z)$ while $\text{NA}(Z, Z)$ is not dense (indeed, $Z = G \oplus_\infty \ell_2$ where G is Gowers' space describe in Example 5.10 would work). We do not know whether such an example exists.

Problem 7.4. Does there exist a Banach space Z such that $\text{QNA}(Z, Z)$ is dense in $\mathcal{L}(Z, Z)$ while $\text{NA}(Z, Z)$ is not dense?

7.2. Lindenstrauss properties. It would be of interest to study the version for quasi norm attainment of Lindenstrauss properties A and B. Let us say that a Banach space X has *property quasi A* if $\overline{\text{QNA}(X, Z)} = \mathcal{L}(X, Z)$ for every Banach space Z . A Banach space X has *property quasi B* if $\overline{\text{QNA}(U, X)} = \mathcal{L}(U, X)$ for every

Proof intuition

For an operator into an ℓ_∞ -sum, quasi norm attainment can fail to patch coordinatewise because one needs a single sequence along which both coordinates converge. There are two useful positive mechanisms. The first is classical: if one coordinate is made strictly dominant and norm attaining, then the whole sum is norm attaining, so the other coordinate needs no hypothesis. The second is genuinely quasi-norm-attaining: if the dominant coordinate is only QNA, compactness of every subordinate coordinate supplies the missing convergent subsequence. In both cases strict dominance ensures that the limit in the product still has the full operator norm.

Main partial result

Theorem 1 (Norm-attaining dominant-coordinate subcase). *Let X, Y_1, Y_2 be Banach spaces and let $T = (T_1, T_2) \in \mathcal{L}(X, Y_1 \oplus_\infty Y_2)$.*

Assume that for some $j \in \{1, 2\}$,

$$\|T_j\| = \|T\| = \max\{\|T_1\|, \|T_2\|\},$$

and that $T_j \in \overline{\text{NA}(X, Y_j)}$. Then

$$T \in \overline{\text{QNA}(X, Y_1 \oplus_\infty Y_2)}.$$

Proof. By symmetry suppose $j = 1$. If $T = 0$, there is nothing to prove; put $M = \|T_1\| = \|T\| > 0$. Fix $\varepsilon > 0$ and choose $0 < \delta < \min\{\varepsilon/10, M/8\}$. Pick $A_1 \in \text{NA}(X, Y_1)$ with $\|A_1 - T_1\| < \delta$. Set $\alpha = 4\delta/M$ and $S_1 = (1 + \alpha)A_1$. Then $S_1 \in \text{NA}(X, Y_1)$,

$$\|S_1 - T_1\| \leq \|A_1 - T_1\| + \alpha\|A_1\| < 6\delta < \varepsilon,$$

and

$$\|S_1\| \geq (1 + \alpha)(M - \delta) > M + 2\delta \geq \|T_2\| + 2\delta.$$

Let $S = (S_1, T_2)$. Then $\|S - T\| < \varepsilon$ and $\|S\| = \|S_1\|$ by strict dominance. Since S_1 attains its norm, choose $x_0 \in S_X$ with $\|S_1 x_0\| = \|S_1\|$. We have

$$\|S x_0\|_\infty = \max\{\|S_1 x_0\|, \|T_2 x_0\|\} = \|S_1\| = \|S\|,$$

so $S \in \text{NA}(X, Y_1 \oplus_\infty Y_2) \subseteq \overline{\text{QNA}(X, Y_1 \oplus_\infty Y_2)}$. □

Theorem 2 (Compact-subordinate ℓ_∞ stability). *Let X, Y_1, Y_2 be Banach spaces and write $Y = Y_1 \oplus_\infty Y_2$. Let $T = (T_1, T_2) \in \mathcal{L}(X, Y)$.*

Assume that for some $j \in \{1, 2\}$,

$$\|T_j\| = \|T\| = \max\{\|T_1\|, \|T_2\|\},$$

that $T_j \in \overline{\text{QNA}(X, Y_j)}$, and that the other coordinate T_k lies in $\overline{\mathcal{K}(X, Y_k)}$, where $\{j, k\} = \{1, 2\}$. Then

$$T \in \overline{\text{QNA}(X, Y_1 \oplus_\infty Y_2)}.$$

Proof. By symmetry suppose $j = 1$ and $k = 2$. If $T = 0$, the conclusion is immediate, so put $M = \|T_1\| = \|T\| > 0$. Fix $\varepsilon > 0$. Choose

$$0 < \delta < \min\{\varepsilon/10, M/8\}.$$

Since $T_1 \in \overline{\text{QNA}(X, Y_1)}$, pick $A_1 \in \text{QNA}(X, Y_1)$ with

$$\|A_1 - T_1\| < \delta.$$

Set $\alpha = 4\delta/M$ and $S_1 = (1 + \alpha)A_1$. Then $S_1 \in \text{QNA}(X, Y_1)$ and

$$\|S_1 - T_1\| \leq \|A_1 - T_1\| + \alpha\|A_1\| < \delta + \frac{4\delta}{M}(M + \delta) < 6\delta.$$

Moreover,

$$\|S_1\| \geq (1 + \alpha)(M - \delta) = M + 3\delta - \frac{4\delta^2}{M} > M + 2\delta.$$

Since $T_2 \in \overline{\mathcal{K}(X, Y_2)}$, choose $K_2 \in \mathcal{K}(X, Y_2)$ with

$$\|K_2 - T_2\| < \delta.$$

Then

$$\|K_2\| \leq \|T_2\| + \delta \leq M + \delta < \|S_1\|.$$

Define $S = (S_1, K_2) \in \mathcal{L}(X, Y_1 \oplus_\infty Y_2)$. We have

$$\|S - T\| = \max\{\|S_1 - T_1\|, \|K_2 - T_2\|\} < 6\delta < \varepsilon,$$

and, by strict dominance, $\|S\| = \|S_1\|$.

It remains to show that S is quasi norm attaining. Since $S_1 \in \text{QNA}(X, Y_1)$, there is a sequence $(x_n) \subset B_X$ and a vector $u_1 \in Y_1$ such that $S_1 x_n \rightarrow u_1$ and $\|u_1\| = \|S_1\|$. The sequence $(K_2 x_n)$ has a norm-convergent subsequence because K_2 is compact; pass to a subsequence and write $K_2 x_n \rightarrow u_2$. Then

$$Sx_n = (S_1 x_n, K_2 x_n) \longrightarrow (u_1, u_2)$$

in $Y_1 \oplus_\infty Y_2$, and

$$\|(u_1, u_2)\|_\infty = \max\{\|u_1\|, \|u_2\|\} = \|S_1\| = \|S\|,$$

since $\|u_2\| \leq \|K_2\| < \|S_1\|$. Hence $S \in \text{QNA}(X, Y)$. As $\varepsilon > 0$ was arbitrary, $T \in \overline{\text{QNA}(X, Y)}$. $\square$

Corollary 1 (Finite ℓ_∞ sums). *Let $Y = Y_1 \oplus_\infty \cdots \oplus_\infty Y_m$ and $T = (T_1, \dots, T_m) \in \mathcal{L}(X, Y)$. Suppose that for some j , $\|T_j\| = \|T\|$, $T_j \in \overline{\text{QNA}(X, Y_j)}$, and $T_i \in \overline{\mathcal{K}(X, Y_i)}$ for every $i \neq j$. Then $T \in \overline{\text{QNA}(X, Y)}$.*

Proof. The proof is the same. Perturb and slightly scale the j th coordinate to be strictly dominant. Approximate every other coordinate by a compact operator. The quasi-norming sequence for the dominant coordinate has a subsequence on which all finitely many compact subordinate coordinates converge. $\square$

Limitations

This is a partial positive result, not a solution of Problem 7.3. The hard case remains when the subordinate coordinate is merely QNA-approximable and no compactness, norm-attainment, or other synchronizing mechanism forces convergence along the dominant coordinate's quasi-norming sequence.

The result also does not settle Problem 7.4 from the same source. In particular, it does not prove that the proposed Gowers-space example $G \oplus_\infty \ell_2$ has dense quasi norm attaining endomorphisms.

Novelty and literature check

The run's cheap indexes had only one previous triage attempt for arXiv:2004.11025 and no durable packet for Problem 7.3. I inspected the source paper and the later paper Choi–Jung–Kim–Martin, *Weak-star quasi norm attaining operators*, arXiv:2209.07763, which treats a weak-star variant and records ordinary QNA separations but does not answer the ordinary ℓ_∞ range sum question. Web searches on 2026-06-19 for exact phrases around “quasi norm attaining” and “ell_infty”, “QNA(X, Y_1)”, and “Problem 7.3 QNA” found the source and the weak-star variant but no later ordinary-QNA resolution.

References

1. G. Choi, Y. S. Choi, M. Jung, and M. Martin, *On quasi norm attaining operators between Banach spaces*, arXiv:2004.11025.
2. G. Choi, M. Jung, S. K. Kim, and M. Martin, *Weak-star quasi norm attaining operators*, arXiv:2209.07763.

Human-review recommendation

Review as a complete partial theorem. The key verification points are the strict-dominance perturbation estimates and the compact-subsequence passage.